

Ensnaring Graphs

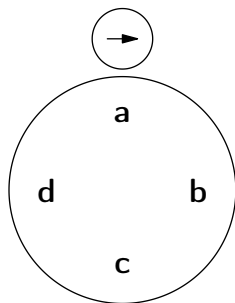
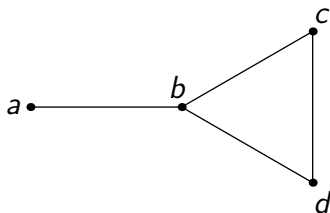
Derek Liu, joint with Colin Defant

REU Duluth

1/10/25

Stone Diagrams

Given a graph, let us associate a label to each vertex. A *stone diagram* consists of these labels in a circle, with a *stone* on one of the labels pointing in either the clockwise or counterclockwise direction.

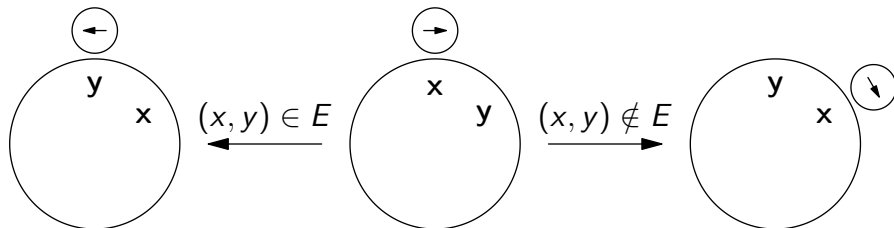


Refractive Toric Promotion

Refractive toric promotion is a process on stone diagrams. Suppose the stone is on x and points towards y .

If x and y are neighbors, the labels swap and the stone turns around.

If x and y aren't neighbors, the labels swap and the stone moves forward.



Contractible Orbits and Ensnaring Graphs

Refractive toric promotion is reversible, so we can consider a stone diagram's orbit.

An orbit is considered *contractible* if every vertex makes a net 0 number of revolutions around the diagram.

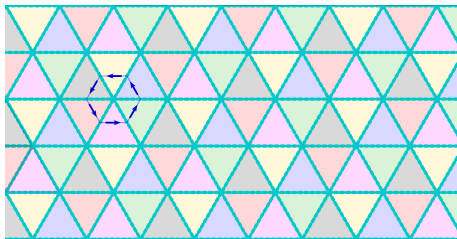
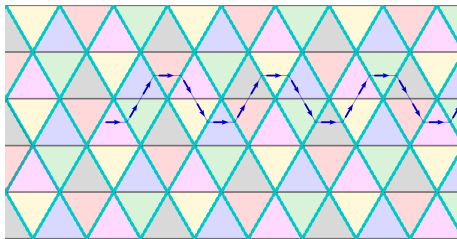
A graph is *ensnaring* if all of its orbits are contractible.

Question

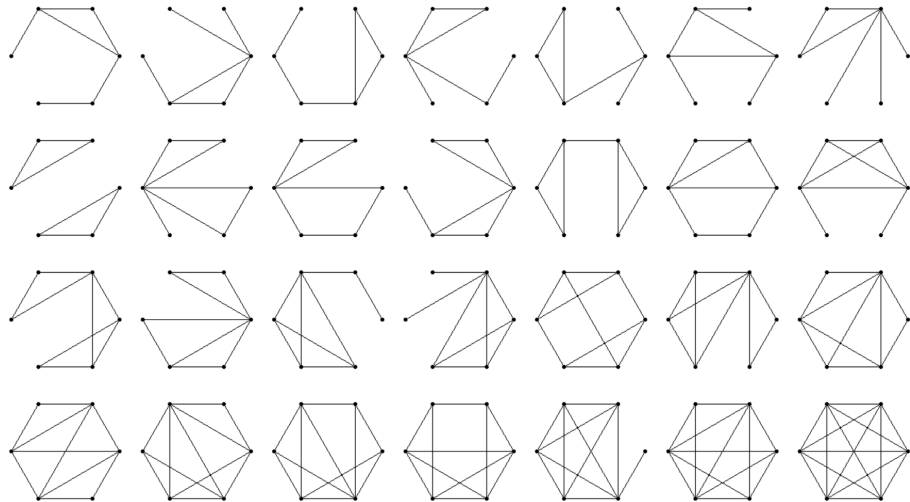
Which graphs are ensnaring?

(A graph is *expelling* if none of its orbits are contractible. These are significantly less interesting; a graph is expelling if and only if it is bipartite.)

Topological Motivation



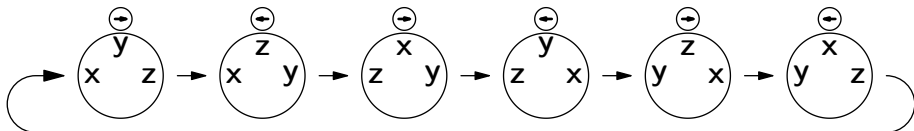
6-vertex Ensnaring Graphs



Complete Graphs and Cycles

Theorem (Defant-L '25+): Complete Graphs are Ensnaring

The complete graph K_n on n vertices is ensnaring as long as $n \geq 3$.

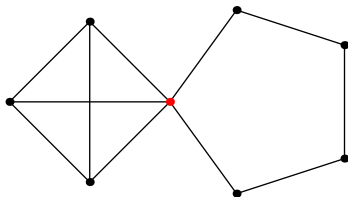


Theorem (Defant-L '25+): Odd Cycles are Ensnaring

The cycle graph C_n on $n \geq 3$ vertices is ensnaring if and only if n is odd.

Wedges

A *wedge* of two graphs is obtained by gluing them together at one vertex.



Theorem (Defant–L '25+): Wedges Preserve Ensnaring

Any wedge of two ensnaring graphs is ensnaring.

More Wedges

The converse of the aforementioned theorem is not true.

Theorem (Defant–L '25+): Wedge of Complete Graph with Trees

Take a complete graph K_n for $n \geq 3$ and wedge a tree onto each vertex. The resulting graph is ensnaring.

Conjecture (Defant–L '25+): Wedges Preserve Non-Ensnaring

The wedge of two non-ensnaring graphs at any vertices cannot be ensnaring.

Disjoint Unions

To determine when disjoint unions of graphs are ensnaring, we need an additional definition.

An ensnaring graph is *revolutionary* if in at least one of the orbits, the stone makes a net nonzero number of revolutions around the diagram.

Theorem (Defant–L '25+): Disjoint Unions

The disjoint union of two graphs is ensnaring if and only if both graphs are ensnaring, but not revolutionary.

Theorem (Defant–L '25+): Most Odd Cycles are Revolutionary

The cycle graph C_n on n vertices is revolutionary for all odd $n \geq 5$.

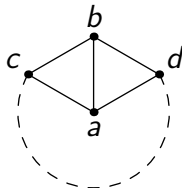
Non-Ensnaing can be a Local Condition

Theorem (Defant–L '25+): Connected Triangles Sharing an Edge

Suppose a graph has vertices a, b, c, d subject to the following:

- a 's neighborhood is exactly $\{b, c, d\}$.
- b 's neighborhood is exactly $\{a, c, d\}$.
- There exists a path from c to d avoiding a and b with at least two edges.

Then the graph is not ensnaing.



Components of the Complement Act Separately

We define the *n -vertex complement* of a graph G to be the complement of G with universal vertices added so that there are n total vertices. We denote this as $\text{compl}_n(G)$.

Theorem (Defant–L '25+): Complement Components Act Separately

A graph G with n vertices is ensnaring if and only if for each connected component C of the complement of G , $\text{compl}_n(C)$ is ensnaring.

Conjecture (Defant–L '25+): Only Parity Matters for Complements

Fix a graph G on m vertices. Then, for all integer $n > m$, whether $\text{compl}_n(G)$ is ensnaring depends only on the parity of n .

Simple Complement Components

Theorem (Defant–L '25+): Complete Graph as Complement

For any integers $n \geq m \geq 2$, $\text{compl}_n(K_m)$ is not ensnaring.

Theorem (Defant–L '25+): Complete Bipartite Graph as Complement

For any integers ℓ , r , and n with $\ell + r < n$, $\text{compl}_n(K_{\ell,r})$ is ensnaring if and only if $n - \ell - r$ is odd and $(\ell, r) \neq (1, 1)$.

Conjecture (Defant–L '25+): Trees in the Complement with Even n

For any even integer n and tree T with $m < n$ vertices, $\text{compl}_n(T)$ is ensnaring if and only if m is odd.

Future Directions

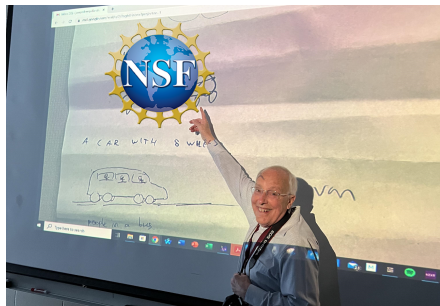
- Resolve the aforementioned conjectures on wedges and complement components
- Understand toric promotion on graphs of middling density
- Identify more revolutionary graphs
- Generalize results to include reflection edges, a different type of edge in the graph which causes stone diagrams to behave differently
- Analyze orbit lengths (especially in graphs which are complements of paths)

Acknowledgements

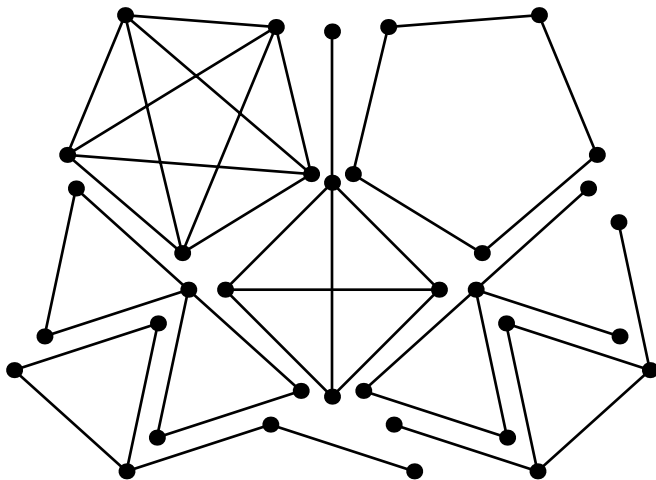
Thanks to advisors Noah Kravitz, Mitchell Lee, and Maya Sankar, as well as visitors, for providing their guidance and support, as well as creating a healthy environment for research!

Thanks to Professor Gallian and Colin Defant for running REU Duluth!

Thanks to Jane Street Capital, NSF, Ray Sidney, and Eric Wepsic for helping fund the program!



Thanks for Listening!



Questions?